BCSC 1006: DATA STRUCTURES AND ALGORITHM

Objective: To introduce fundamental data structures and algorithmic problem solving using Java. The course enables students to understand data organization, analyze algorithm efficiency, and implement solutions using linear data structures, recursion, and hashing for real-world applications.

Credits: 03 **L–T–P: 1–0–4**

Module No.	Content	Teaching Hours
I	**Introduction to Data Structures and Algorithms:** Basic terminology, elementary data organization, data and information, abstract data type (ADT), characteristics and properties of algorithms, problem solving strategies, algorithmic efficiency including time and space complexity, and asymptotic notations such as Big-O, Omega and Theta (introductory level). **Arrays:** Introduction to arrays, memory representation, indexing, traversal, insertion, deletion and updation, linear search, binary search, prefix sum, frequency counting, two pointer technique, sliding window basics, Kadane's algorithm and problem solving patterns on arrays. **Strings**: Strings versus character arrays, mutable and immutable strings, string handling in Java, ASCII and Unicode, character operations, StringBuilder and StringBuffer, hashing in strings, anagram logic, substring and subsequence, string problem solving patterns and sliding window techniques. **Searching and Sorting Algorithms** Searching: Linear Search, Binary Search, Applications of Searching Sorting: Bubble Sort, Selection Sort, Insertion Sort, Comparison of Sorting Algorithms	20
II	**Recursion**: Introduction to recursion, base and recursive cases, recursion tree, stack memory concept, recursion versus iteration, tail recursion, multiple recursion, recurrence thinking (introductory), subsequence generation, pick and not-pick technique. Merge Sort, Quick Sort. **Limitations of arrays including fixed size**, inefficient memory usage, costly insertions and deletions and homogeneous data handling. **Linked Lists:** Introduction, node structure, dynamic memory allocation, traversal, insertion, deletion and searching operations in singly, doubly and circular linked lists, fast and slow pointer technique, cycle detection and dummy node concept. **Stack**: Stack as an abstract data type, implementation using array and linked list, operations such as push, pop and peek, and applications including expression evaluation and recursion simulation. **Queue**: Queue as an abstract data type, implementation using array and linked list, circular queue, deque and introductory priority queue, along with applications such as queue reversal and basic BFS-based problems.	20

Text Book:

"Data Structures and Algorithms in Java" by Michael T. Goodrich, Roberto Tamassia, 6th Edition, Wiley.

Reference Books:

- "Data Structures and Algorithms Made Easy in Java" by Narasimha Karumanchi.
- "Introduction to Algorithms" by Cormen, Leiserson, Rivest, Stein (CLRS) – Java applications referenced.
- "Problem Solving with Algorithms and Data Structures Using Java" by Bradley N. Miller, David L. Ranum.

Outcomes:

- **CO1:** Analyze and evaluate algorithm efficiency using **time and space complexity concepts** and asymptotic notations.
- **CO2:** Apply appropriate techniques on **arrays and strings** such as searching, sorting, two-pointer and sliding window methods to solve computational problems.
- **CO3:** Design and implement solutions using **recursion and divide-and-conquer techniques**.
- **CO4:** Implement and analyze **linked lists, stacks, and queues** using object-oriented programming in Java.
- **CO5:** Select and apply suitable **data structures and problem-solving strategies** to solve real-world and coding problems effectively.

CO–PO MAPPING (STANDARD – NBA STYLE)

Program Outcomes (POs) (typical engineering mapping)

PO Code	Description
PO1	Engineering Knowledge
PO2	Problem Analysis
PO3	Design / Development of Solutions
PO4	Investigation of Complex Problems
PO5	Modern Tool Usage
PO6	Engineer and Society
PO7	Environment & Sustainability
PO8	Ethics
PO9	Individual & Team Work
PO10	Communication
PO11	Project Management
PO12	Life-long Learning

CO–PO Mapping Table (DSA-I)

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	2	1	-	-	-	-	-	-	2
CO2	3	3	2	2	1	-	-	-	-	-	-	2
CO3	3	3	3	2	2	-	-	-	-	-	-	2
CO4	3	3	3	2	2	-	-	-	1	-	-	2
CO5	3	3	3	3	2	-	-	-	1	1	-	3

Mapping Key

- **3 → Strong**
- **2 → Moderate**
- **1 → Low**
- **– → Not mapped**

CO–PSO Mapping

CO / PSO	PSO1 (Programming & Problem Solving)	PSO2 (Software Development)	PSO3 (Emerging Technologies)
CO1	3	2	1
CO2	3	2	1
CO3	3	2	1
CO4	3	3	1
CO5	3	3	2